

5. (a) Two equations of accelerated motion are $v = u + at$ and $v^2 = u^2 + 2ax$. Use these equations to show that, for a body accelerating uniformly from rest: [2]

$$x = \frac{1}{2} at^2$$

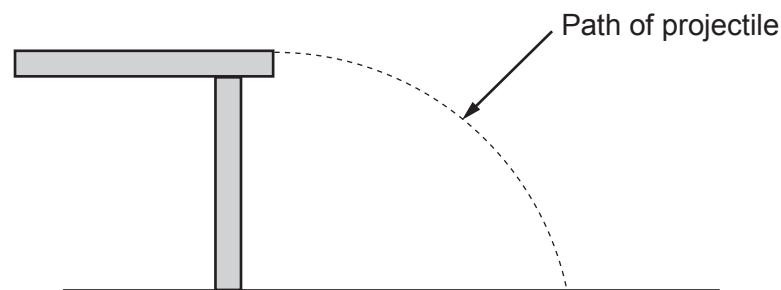
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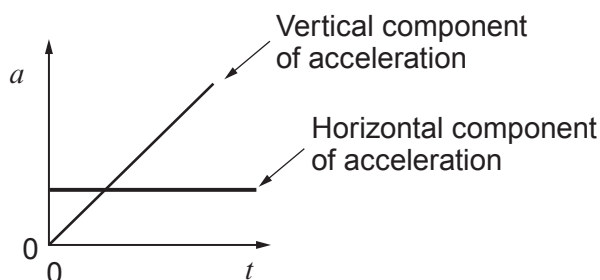
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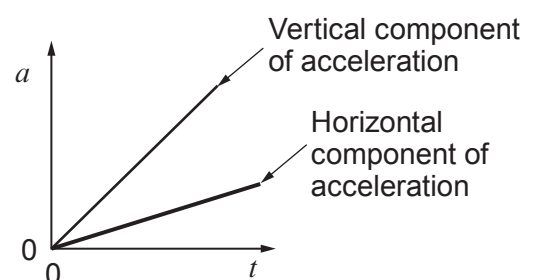
- (b) The diagram shows the path of a projectile after it is launched horizontally from a table.



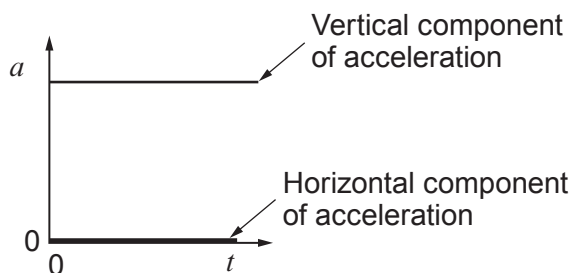
When considering the horizontal and vertical components of the motion, graphs of horizontal acceleration and vertical acceleration against time are sketched. **Only one** of the following sketch graphs shows a correct combination. [Ignore air resistance for the remainder of the question.]



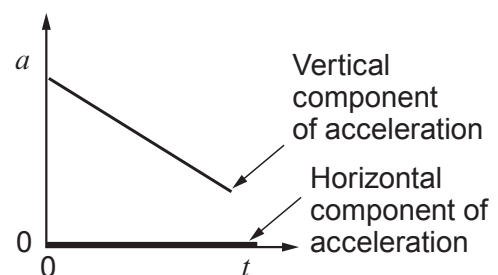
Graph 1



Graph 2



Graph 3



Graph 4



State which graph shows the correct combination and explain your answer.

[3]

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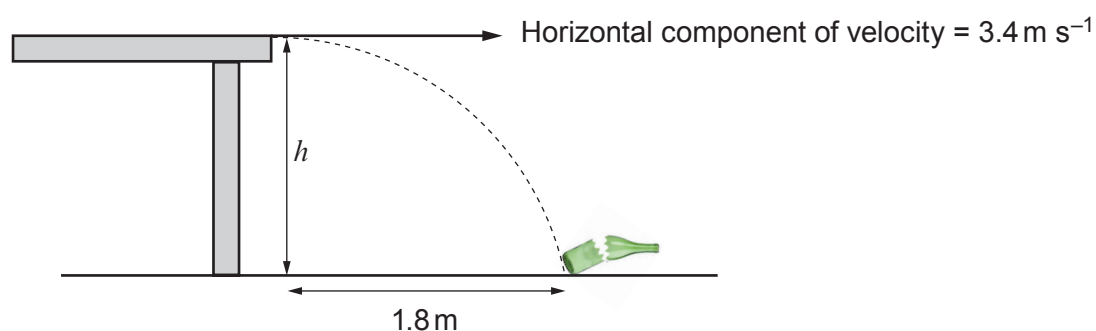
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(c) A bottle is accidentally knocked from the table and follows the path shown.



(i) Calculate the height, h , of the table.

[3]

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- (ii) Calculate the magnitude of the velocity and the direction of travel of the bottle just before it hits the ground. [4]

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- (d) State whether or not the following statement is correct and justify your answer. [2]

The flight time for the bottle in part (c) will depend on the horizontal velocity – the greater the horizontal velocity, the longer it will take for the bottle to hit the floor after leaving the table.

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